Precalculus

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1 Introduction

These notes go over "precalculus," the material at the high school level up to but not including calculus. Since things move a lot more quickly and less time is given to routine mechanical processes, it's maybe better seen as a rigorous, second-pass-through precalculus text to prepare for calculus.

Remark 1.0.1. When we introduce a function, we specify its domain and range (each being sets, respectively) by the notation

$$f : A \to B$$

where f is the function name, A is the domain, and B is the codomain. Remember that functions are little "machines" that take in objects of the domain A and produce a unique object in the codomain B.

When we want to specify a function that takes in more than one argument, we have the domain be a *Cartesian product*, which allows for bundling of information—as an example, the pair of real numbers (a, b) belongs not to $\mathbb{R}$ but to $\mathbb{R} \times \mathbb{R}$ or $\mathbb{R}^2$. Thus, addition is considered a function $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$, taking in two numbers and producing a sum.

To say an object a is in a set A (i.e., a is an *element* or *member* of A), we write $a \in A$. If every element of A is also contained in a set B, we say A is a *subset* of B and write $A \subseteq B$.

2 Real numbers

The *real numbers* describe "everything on the number line," so all of the whole numbers plus everything in-between. We denote by $\mathbb{R}$ the set of real numbers, or the number line. We won't lay out the precise mathematical formulation of the real numbers, because that's a bit technical, but generally you should assume everything works as expected with respect to the basic math operations $+, -, \times, \div$ and comparisons $<, \leq, =, \geq, >$. If you are interested in a formal treatment, consult a real analysis textbook.

2.1 Absolute value and inequalities

Definition 2.1.1. The *absolute value* $|-|:\mathbb{R}\to\mathbb{R}$ forces a number into its positive form:

$$|x| = \begin{cases} x & x \text{ positive}, \\ -x & x \text{ negative}. \end{cases}$$

Interpret the above expression as "If x is positive, then $|x| = x$. If x is negative, then $|x| = -x$." This ensures that $|x|$ is positive no matter what.

Proposition 2.1.2. *Some obvious facts:*

$$|x| \geq 0, \qquad |-x| = |x|, \qquad |xy| = |x||y|.$$

Proposition 2.1.3 (triangle inequality)**.** $|x+y| \leq |x| + |y|$

Proof. We know $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$. Adding these inequalities together gives

$$-(|a| + |b|) \leq a + b \leq |a| + |b|. \tag{1}$$

But remember that $-c \leq x \leq c$ for positive c in general is equivalent to saying $|x| \leq c$, so (1) is equivalent to $|a+b| \leq |a| + |b|$. □

Definition 2.1.4. Given real numbers a and b, we denote by (a,b) the *open interval* from a to b. A number $x \in (a,b)$ is in the interval if and only if $a < x < b$. The *closed interval* $[a,b]$ includes the endpoints: $x \in [a,b]$ if and only if $a \leq x \leq b$. You can also have $(a,b]$ and $[a,b)$ which are defined as expected.

Proposition 2.1.5. *Suppose $c > 0$. Then $|x| < c$ is equivalent to saying $x \in (-c,c)$, and $|x| \leq c$ is equivalent to saying $x \in [-c,c]$.*

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